Design of an Energy-Harvester-Powered Power Management Unit for Low-Power Applications Using 0.18 µm CMOS Technology in Cadence Virtuoso

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***Abstract*—Battery-less sensor nodes and Internet of Things (IoT) devices depend on energy harvested from the environment, but harvested sources such as small solar cells deliver low and fluctuating voltages, approximately 0.3–1 V, which cannot power electronic loads directly. A power management unit (PMU) is therefore required to condition, store and regulate the harvested energy. This report presents the architecture of a proposed solar energy-harvesting PMU and the design and standalone characterization of its first implemented block: a two-stage CMOS operational amplifier intended as the error amplifier of a future low-dropout (LDO) regulator. The amplifier uses a PMOS differential input pair with a current-mirror load, a common-source second stage, and Miller compensation with a series nulling resistor. It was designed by square-law hand sizing and simulated in Cadence Virtuoso using the gpdk180 process at a 1.8 V supply and a 0.5 V input common mode. The simulated amplifier achieves approximately 70.5 dB open-loop gain, 1.629 MHz unity-gain bandwidth, 51.8° phase margin, 87.8 dB CMRR at 10 Hz, and a reported ADE power figure of 4.73 µW at 3.27 µA supply current. The phase margin falls short of the 60° design guideline, and the complete PMU — LDO, reference, boost/conditioning stage and voltage monitor — remains proposed rather than implemented. The report separates verified amplifier results from proposed LDO values and sets out the verification plan for the remaining blocks.**

***Index Terms*—Energy harvesting, power management unit, low-dropout regulator, error amplifier, two-stage CMOS operational amplifier, Miller compensation, 0.18 µm CMOS.**

I. INTRODUCTION

Wireless sensor networks and IoT deployments increasingly favour nodes that operate without battery replacement, which places the burden of supplying power on energy harvested from the environment. Among the available ambient sources, small solar cells are attractive for such nodes because of their comparatively high power density and their natural compatibility with CMOS integration; this motivated the solar-harvesting PMU of Patel et al., which regulates the output of on-chip solar cells using a 0.18 µm CMOS LDO and serves as the principal reference for this project [1].

Harvested electrical energy is, however, poorly suited to direct use. The project presentation models two series solar cells that deliver roughly 0.3–1 V depending on illumination, and the available power varies with the environment. A PMU bridges this gap: it conditions the input, stores energy, generates a reference, regulates the supply seen by the load, and can disconnect the load when stored energy is insufficient. Within such a PMU, the regulation duty falls on a linear LDO stage, whose accuracy and stability are set largely by its error amplifier [2], [3]. The error amplifier compares a divided copy of the output with a reference and drives the gate of a PMOS pass transistor, so its gain, bandwidth, phase margin and quiescent current directly determine the regulation quality and the power overhead of the whole unit.

This project therefore begins the PMU design with its most performance-critical analog block. A two-stage CMOS operational amplifier was designed by square-law hand analysis, implemented in Cadence Virtuoso on the generic 0.18 µm process design kit (gpdk180), and characterized with the Spectre simulator. The 0.18 µm node was selected because it is the technology of the reference PMU [1], offers well-characterized 1.8 V devices with adequate headroom for a two-stage topology, and is the process available in the institute's Cadence environment. This report documents the architecture, sizing, simulation methodology and measured standalone results of the amplifier, and presents the proposed — not yet implemented — LDO and PMU integration plan.

II. PROBLEM STATEMENT, OBJECTIVES AND SCOPE

Problem statement. To design an energy harvester-powered power management unit for low-power applications using 0.18 µm CMOS technology in Cadence Virtuoso, converting a low and fluctuating harvested input into a regulated supply suitable for low-power loads such as sensor nodes.

The measurable objectives, with their verification methods and present status, are collected in Table I. Objectives 1–3 concern the error amplifier and are complete at schematic level; objectives 4–7 define the remaining system work. The scope completed so far is the standalone amplifier: schematic entry, DC operating-point verification, AC gain/phase, CMRR, transient and slew-rate simulation. The proposed scope — LDO closure around a PMOS pass device, reference, input conditioning or boost stage, storage, and voltage monitoring — is specified but unverified.

TABLE I

PROJECT OBJECTIVES AND VERIFICATION CRITERIA

#	Objective	Requirement	Verification	Status
1	Design two-stage error amplifier in gpdk180	All devices saturated at V_{DD}=1.8 V, V_{ICM}=0.5 V	DC operating point	Done (all region 2)
2	Meet amplifier small-signal targets	A_v≥60 dB; GBW 10–2000 kHz; CMRR≥80 dB	AC / CMRR analyses	Done (70.5 dB; 1.629 MHz; 87.8 dB)
3	Stability and dynamics	PM≥60°; SR≥0.1 V/µs; P<10–100 µW	AC phase, transient, ADE power	Partial — PM 51.8° below 60°
4	Close LDO loop (PMOS pass + divider)	V_{out}=1.2 V from V_{ref}=0.5 V at ≤100 µA	DC, load/line sweeps, loop gain	Proposed — not simulated
5	Voltage reference	0.5 V, low-power	DC, temperature sweep	Proposed
6	Harvest interface / boost + storage	Operate from 0.3–1 V source	System transient	Proposed
7	Voltage monitor and load control	Connect load only when energy suffices	System transient	Proposed

III. RELATED WORK

Patel, Mishra and Nagchoudhuri integrate on-chip solar cells with a 36 nW PMU in 0.18 µm CMOS, using a subthreshold cascoded error amplifier inside an LDO; their work demonstrates the feasibility of full integration but accepts modest amplifier gain (≈46 dB) and bandwidth as the price of nanowatt operation [1]. The present project deliberately diverges by biasing the amplifier in strong/moderate inversion from a 1.8 V rail, trading microwatt quiescent power for higher, more predictable gain. Rincón-Mora and Allen established the canonical low-quiescent-current LDO analysis — pass-device sizing, loop stability with the output capacitor and its equivalent series resistance (ESR) — that the proposed LDO stage follows [2]. For the amplifier itself, the two-stage Miller-compensated topology and the nulling-resistor technique for cancelling the right-half-plane (RHP) zero are standard results covered in Razavi and in Gray et al. [3], [4]; Allen and Holberg give the square-law design procedure used for hand sizing [5]. On the system side, Seok et al. demonstrate a picowatt 2-transistor 0.5 V reference suitable for harvesting PMUs [6], and Seeman and Sanders provide the analysis framework for the switched-capacitor conversion stage that the 0.3–1 V source will require [7]. Table II compares these works; NR marks values not reported by the cited source.

TABLE II

LITERATURE COMPARISON (NR = NOT REPORTED IN THE CITED SOURCE)

Ref.	Tech.	Architecture	Supply	Power	Gain / BW	Contribution	Limitation
[1]	0.18 µm	Solar PMU: LDO + subthreshold cascoded error amp	≈0.98 –1.33 V in	36 nW (PMU)	46.39 dB / 206 kHz UGB	Fully integrated nW solar PMU	Subthreshold: low gain, slow
[2]	2 µm BiCMOS	Low-dropout regulator	NR	NR	NR	LDO stability/quiescent analysis	Older node; discrete C_{out}
[6]	0.13 µm	2-transistor voltage reference	0.5 V	picowatt	—	Ultra-low-power 0.5 V reference	Untrimmed accuracy limits
[7]	— (theory)	Switched-capacitor DC–DC analysis	—	—	—	SC converter optimization method	No silicon demonstrator
This work	0.18 µm (gpdk180)	Two-stage error amp for PMU LDO	1.8 V	4.73 µW (ADE, def. pending)	70.5 dB / 1.629 MHz	Verified standalone error amp	PM 51.8°; system blocks proposed only

IV. PROPOSED PMU ARCHITECTURE

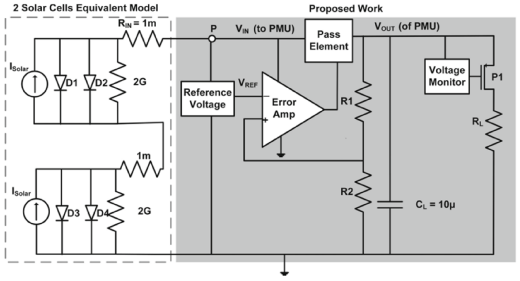


Fig. 1. Proposed PMU block diagram from the project presentation: solar-cell equivalent model (left) feeding the proposed regulation chain — reference, error amplifier, pass element, R1/R2 feedback divider, C_L = 10 µF output capacitor, voltage monitor and load switch P1. Only the error amplifier is implemented to date.

Energy and signal flow proceed left to right in Fig. 1. The harvester (modelled as two series solar cells, each an I_{solar} source with diodes and a 2 GΩ shunt behind a 1 mΩ series resistance) feeds the PMU input node. A reference generator derives V_{ref}; the error amplifier compares V_{ref} with the feedback voltage tapped from the R1/R2 divider and drives the pass element so that the output settles where both inputs balance. The output capacitor holds V_{qut} between energy arrivals, and a voltage monitor gates the load through switch P1 so the node draws power only when the stored energy permits — the duty-cycled operating style of harvesting PMUs [1].

Voltage headroom. A 0.3–1 V source cannot drive a conventional LDO whose target output is 1.2 V: a linear regulator only drops voltage, and it needs an input above the output by at least the pass-device dropout. The architecture therefore requires either a boost or switched-capacitor conversion stage between the harvester and the LDO input, together with an energy-storage element that accumulates charge toward the ≈1.8 V rail assumed for the amplifier characterization [7]. This conditioning stage is part of the proposed scope; the

amplifier results reported here were obtained from an ideal 1.8 V supply and do not demonstrate operation from the harvester. Table III summarizes the status of every block.

TABLE III
SYSTEM-BLOCK STATUS

PMU block	Function	Status	Next action
Error amplifier	Compare V_{fb} with V_{ref}, drive pass gate	Implemented, verified	Re-verify inside closed LDO loop
PMOS pass element	Drop V_{IN}–V_{out}, source load current	Proposed (60 µm/0.5 µm)	DC + load-sweep verification
Feedback divider R1/R2	Scale V_{out} to V_{fb}	Proposed (700 kΩ/500 kΩ)	Confirm ratio, current draw
Voltage reference	0.5 V set-point	Proposed	Select & simulate topology [6]
Boost / conditioning + storage	Raise 0.3–1 V source to LDO input; buffer energy	Proposed	Topology study [7], sizing
Voltage monitor + P1	Load connect/disconnect	Proposed	Threshold design, hysteresis

V. TWO-STAGE ERROR-AMPLIFIER DESIGN

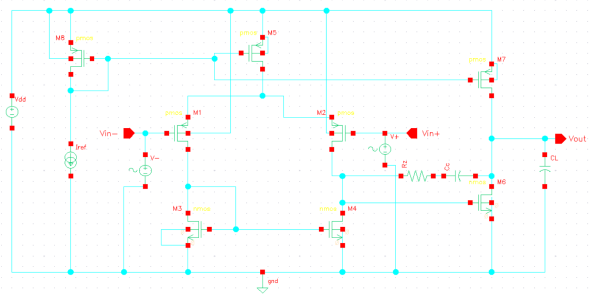


Fig. 2. Cadence Virtuoso schematic of the two-stage amplifier (gpdk180): PMOS input pair M1/M2, NMOS mirror load M3/M4, PMOS bias mirror M8/M5/M7 from I_{ref}, common-source second stage M6 with PMOS load M7, and R_Z–C_C Miller network from the first-stage output to V_{out}; C_L models the load.

The input pair M1/M2 is PMOS. With a 0.5 V common mode an NMOS pair would not remain in saturation — its required minimum common mode (tail overdrive plus V_{TH} plus input overdrive) exceeds 0.5 V — whereas a PMOS pair referenced to the 1.8 V rail senses low common-mode inputs naturally; PMOS inputs also exhibit lower 1/f noise, which matters for the low-frequency biomedical-class loads the project targets [3]. The NMOS mirror M3/M4 converts the differential current to a single-ended signal at the M4 drain; M8 is the diode-connected master of the PMOS bias mirror fed by I_{ref}, with M5 supplying the 1 µA tail and M7 the second-stage load current. M6 forms a common-source second stage, giving the output swing needed to drive a PMOS pass gate. Stability is set by pole splitting through C_C with series resistor $R_Z \approx 1/g_{m6}$ to cancel the RHP zero [4], [5]. The exact C_C, R_Z and C_L component values are not visible in the supplied schematic captures and must be confirmed from the schematic properties: [INSERT VERIFIED $C_C/R_Z/C_L$ FROM CADENCE].

A. Hand sizing

Devices were sized with the square-law expression, using the first-order gpdk180 stand-in constants $k_n = \mu_n C_{ox} = 316$ µA/V² and $k_p = 62.5$ µA/V² taken from the reference paper's device table [1]:

$$W/L = 2\,I_D / (k \cdot V_{ov}^{\,2}) \quad (1)$$

where I_D is the branch drain current and $V_{ov} = V_{GS} - V_{TH}$ the chosen overdrive. Overdrives of 0.12 V (input pair, maximizing g_m/I_D), 0.15 V (NMOS mirror and M6) and 0.20 V (PMOS current sources) give the sizing of Table IV. Equation (1) is an idealized long-channel approximation: the final operating point must come from simulation, and Section VII quantifies where it deviates.

TABLE IV
TRANSISTOR SIZING (FROM THE PROJECT'S CALCULATED TABLE)

Device	Type	I_D	V_{ov}	W/L calc	W	L	Sizing rationale
M1, M2	PMOS	0.5 µA	0.12 V	1.11	1.1 µm	1 µm	Input pair; low V_{ov} for g_m/I_D
M3, M4	NMOS	0.5 µA	0.15 V	0.14	0.4 µm	2.855 µm	Mirror load; min-width bound, long L for matching/r_o
M5	PMOS	1.0 µA	0.20 V	0.80	0.8 µm	1 µm	Tail source (mirrors M8 ×1)
M6	NMOS	2.0 µA	0.15 V	0.56	0.56 µm	1 µm	2nd-stage CS; matches M3/M4 V_{ov} for offset
M7	PMOS	2.0 µA	0.20 V	1.60	1.6 µm	1 µm	2nd-stage load (mirrors M8 ×2)
M8	PMOS	1.0 µA	0.20 V	0.80	0.8 µm	1 µm	Bias diode from I_{ref} = 1 µA

B. Small-signal design relations

The first-stage gain, second-stage gain, unity-gain bandwidth and slew rate obey the standard two-stage relations [4], [5]:

$$A_{v1} = g_{m1}\,(r_{o2} \,||\, r_{o4}) \quad (2)$$
$$A_{v2} = g_{m6}\,(r_{o6} \,||\, r_{o7}) \quad (3)$$
$$GBW = g_{m1} / (2\pi C_C) \quad (4)$$
$$SR = I_{tail} / C_C \quad (5)$$

where g_{mi} and r_{oi} are the transconductance and output resistance of device M_i, C_C the Miller capacitor and I_{tail} the tail current. With the simulated $g_{m1} = 6.43$ µS and the measured GBW of 1.629 MHz, (4) implies an effective compensation capacitance of about 0.63 pF; this is a consistency check derived from measured values, not a design entry, and the drawn C_C must still be read from the schematic.

VI. SIMULATION METHODOLOGY

All simulations ran in the Cadence Analog Design Environment (ADE) with the Spectre engine on gpdk180 models, $V_{DD} = 1.8$ V, $I_{ref} = 1$ µA and 27 °C, as evidenced by the ADE session captures. The recorded analyses are: (i) a DC operating point with on-schematic annotation to confirm every device in saturation (region 2); (ii) an AC sweep (10 Hz–1 GHz) of the differential gain for DC gain, −3 dB bandwidth and unity-gain frequency; (iii) phase of the same response at the unity-gain frequency for phase margin; (iv) a CMRR measurement computed as the ratio of differential to common-mode gain, $\mathrm{dB20}[(V_{out}/V_{in+})/(V_{out,cm}/V_{cm})]$, using a replica bench driven by a common-mode source; (v) a transient run (10 ms, ±1 mV differential sine at the 0.5 V common mode) and a 2 ms transient from which ADE's slewRate function reports 611.877 kV/s; and (vi) an ADE output named Power that reports 4.7345 µW. The expression behind that Power output is not visible in the captures, so its definition — and in particular whether it averages over a supply sweep that was active in the

same session — requires confirmation: [CONFIRM ADE POWER EXPRESSION]. Analyses not evidenced (PSRR, noise, corners, Monte Carlo, post-layout) were not performed. Closing the LDO will additionally require input-line and load sweeps, loop-gain (stability) analysis with the 10 µF output capacitor and its ESR, start-up transients and PSRR [2].

VII. RESULTS AND DISCUSSION

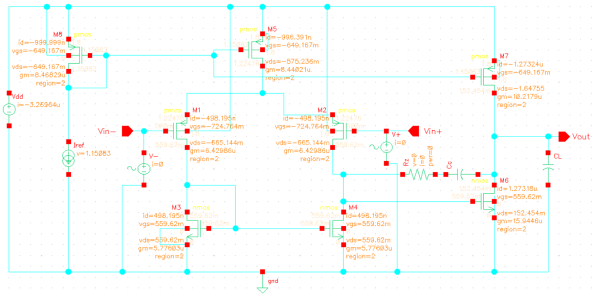


Fig. 3. DC operating point annotated on the schematic: every device reports region 2 (saturation); supply current i(V_{DD}) = 3.26964 µA; tail devices carry ≈0.996 µA and the second stage 1.273 µA.

A. Operating point

Table V extracts the annotated operating point. All eight transistors sit in saturation, validating the bias chain: M8 and M5 mirror I_{ref} accurately (0.9964–1.0 µA) and the pair splits the tail evenly (0.498 µA each). Two deviations from the hand design deserve note. First, the second-stage current is 1.273 µA rather than the intended 2 µA (a –36 % deviation), so the total supply current is 3.27 µA instead of the planned 4 µA; the $W_7/W_8 = 2$ mirror ratio is not realized exactly at this operating point, which first-order sizing with assumed k_p and channel-length modulation at $|V_{DS}|$ = 1.65 V readily explains. Second, the open-loop output node rests at approximately 0.15 V (V_{DS} of M6), far from mid-rail — expected for an uncommitted high-gain amplifier, and a reminder that this bench fixes the bias only; in the LDO the feedback loop will place the output where the pass device requires.

TABLE V
DC OPERATING POINT (FROM THE ANNOTATED SCHEMATIC, FIG. 3)

Device	I_D	V_{GS}	V_{DS}	g_m	Region	Interpretation
M1/M2	–0.498 µA	–724.8 mV	–665.1 mV	6.43 µS	2 (sat)	Balanced input pair
M3/M4	0.498 µA	559.6 mV	559.6 mV	5.78 µS	2 (sat)	1:1 mirror load, diode-matched
M5	–0.996 µA	–649.2 mV	–575.2 mV	8.44 µS	2 (sat)	Tail ≈ I_{ref}
M6	1.273 µA	559.6 mV	152.5 mV	15.94 µS	2 (sat)	CS stage; output rests low (open loop)
M7	–1.273 µA	–649.2 mV	–1.648 V	10.22 µS	2 (sat)	36 % below 2 µA design intent
M8	–1.000 µA	–649.2 mV	–649.2 mV	8.47 µS	2 (sat)	Bias diode; V(I_{ref})=1.151 V

B. Frequency response, stability, CMRR

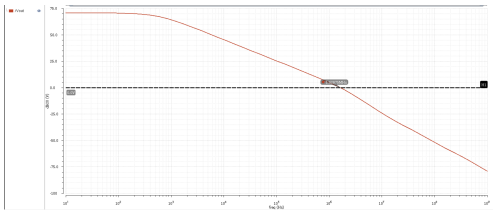


Fig. 4. Open-loop magnitude response: ≈70.5 dB low-frequency gain; marker at the 0 dB crossing gives a unity-gain bandwidth of 1.62875 MHz. A separate marker at gain–3 dB (67.5 dB) gives the –3 dB bandwidth of 550.915 Hz.

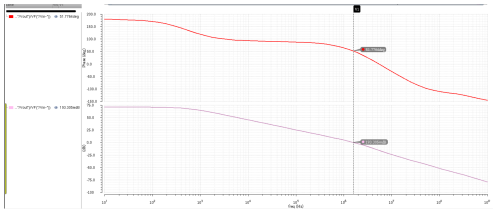


Fig. 5. Phase (top) and magnitude (bottom) at the unity-gain frequency: the phase marker reads 51.7764° where the gain is 0.193 dB, i.e. a phase margin of ≈51.8°.

The amplifier reaches 70.5 dB of DC gain — inside the 60–80 dB target band — with a single dominant pole at ≈551 Hz and a 0 dB crossing at 1.629 MHz, consistent with a gain-bandwidth product of $A_v \cdot f_{-3dB} \approx 3350 \times 551$ Hz ≈ 1.85 MHz to first order. The phase margin is approximately 51.8°. This is a genuine shortfall against the 60° guideline adopted at specification time (the presentation's final table relaxes the target to 45–60°): 51.8° is stable and acceptable for initial characterization, but a two-stage amplifier destined for an LDO loop — where the pass device and output capacitor add further phase lag — should enter that loop with margin in hand. Increasing C_C, retuning R_Z toward $1/g_{m6}$, or accepting a lower GBW are the standard remedies [4]; this retuning is scheduled before loop closure rather than claimed complete.

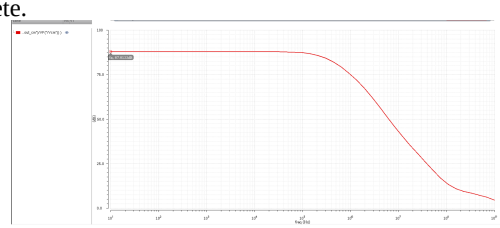


Fig. 6. CMRR versus frequency computed as dB20[(V_{out}/V_{in+})/($V_{out,cm}$/V_{cm})]: 87.8133 dB at 10 Hz, flat through the audio band and rolling off beyond ≈100 kHz.

C. Transient behaviour and power

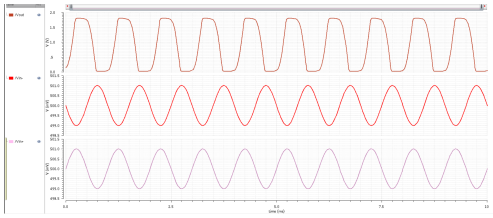


Fig. 7. Open-loop transient with a ±1 mV, ≈ 1 kHz differential drive about the 0.5 V common mode: the 70.5 dB gain saturates the output, which swings essentially rail to rail (≈0.05–1.8 V).

Fig. 7 is an open-loop test: a millivolt-level differential input multiplied by ≈3350 V/V exceeds the supply, so the output clips and the waveform mainly demonstrates polarity, swing and the absence of latch-up — it is not an LDO load-step response and must not be read as one. The ADE slewRate output on a dedicated 2 ms transient reports 611.877 kV/s ≈ 0.612 V/µs,

meeting the ≥0.1 V/µs target. For power, two numbers coexist in the evidence and must not be conflated: the ADE output reports 4.7345 µW (definition unconfirmed, Section VI), while the DC operating point implies

$$P = V_{DD} \cdot I_{DD} = 1.8\ V \times 3.26964\ \mu A \approx 5.885\ \mu W \quad (6)$$

Both lie comfortably inside the <10–100 µW budget, but the ≈20 % gap between them is a measurement-definition question, not a physical result, and is flagged for resolution. Table VI assembles the verified results.

TABLE VI
AMPLIFIER SIMULATION RESULTS (STANDALONE, gpdk180, 1.8 V)

Parameter	Target	Simulated	Unit	Assessment	Evidence
DC open-loop gain	60–80	≈70.5	dB	Met	Fig. 4
–3 dB bandwidth	—	550.915	Hz	Consistent w/ GBW	Fig. 4 marker
Unity-gain bandwidth	0.01–2	1.62875	MHz	Met	Fig. 4
Phase margin	≥60 (relaxed 45–60)	≈51.8	°	Below 60° — retune before LDO	Fig. 5
CMRR @ 10 Hz	80–100	87.8133	dB	Met	Fig. 6
Slew rate	≥0.1–1	≈0.612	V/µs	Met	ADE slewRate output
Supply current (DC op)	—	3.26964	µA	—	Fig. 3
Power (ADE output)	<10–100	4.7345	µW	Met; definition pending	ADE session capture
Power ($V_{DD}\cdot I_{DD}$)	<10–100	≈5.885	µW	Met; from (6)	Fig. 3
PSRR, noise, corners	—	Not measured	—	Pending	—

VIII. PROPOSED LDO IMPLEMENTATION AND VERIFICATION PLAN

The amplifier connects into the LDO without internal modification: its output drives the gate of an external PMOS pass transistor whose source ties to the LDO input rail, V_{ref} = 0.5 V drives the inverting input, and the non-inverting input takes the feedback voltage from the R1/R2 divider across the output. The divider sets the regulated output as

$$V_{out} = V_{ref}(1 + R_1/R_2) \quad (7)$$

so the proposed R_1 = 700 kΩ and R_2 = 500 kΩ with a 0.5 V reference give V_{out} = 0.5(1+1.4) = 1.2 V, with ≈1 µA standing divider current. The remaining component proposals from the LDO implementation manual are listed in Table VII; all are starting values for simulation, none is a verified result. Note that the LDO implementation manual itself was not present in the supplied evidence folder; its values are reproduced from the project brief and must be cross-checked against the manual before submission. Table VIII defines the verification matrix the LDO must pass; the large 10 µF output capacitor and its 16 Ω ESR dominate loop dynamics and will interact with the amplifier's 51.8° margin, which is precisely why loop-gain simulation precedes any performance claim [2].

TABLE VII
PROPOSED LDO COMPONENTS (STARTING VALUES — NOT SIMULATED RESULTS)

Component / parameter	Proposed value	Purpose	Verification status
Nominal LDO input	1.8 V	Supply during initial closure	Pending
Target output	1.2 V	Regulated rail for loads	Pending
Reference voltage	0.5 V	Set-point for (7)	Pending (block proposed)
Max load current	100 µA	Design load envelope	Pending
PMOS pass device	W = 60 µm, L = 0.5 µm	Sources load at low dropout	Pending (ratings, region)
R_1 / R_2	700 kΩ / 500 kΩ	Feedback divider, (7)	Pending
Output capacitor	10 µF	Hold-up, dominant load pole	Pending (realizability)
Capacitor ESR	16 Ω	ESR zero for compensation	Pending

TABLE VIII
LDO VERIFICATION MATRIX (ALL ITEMS CURRENTLY: NOT MEASURED)

Test	Conditions	Measured quantity	Acceptance criterion
DC closure	V_{IN}=1.8 V, no load	V_{out}, device regions	1.2 V ± tol.; all devices saturated
Line sweep	V_{IN} swept about 1.8 V	$\Delta V_{out}/\Delta V_{IN}$	Line regulation within spec to be set
Load sweep	0→100 µA	$\Delta V_{out}/\Delta I_L$	Load regulation within spec; dropout noted
Loop gain / stability	With C_{out}=10 µF, ESR 16 Ω	PM, GM of the loop	PM ≥ 45° over load range
Load/line step	Step I_L and V_{IN}	Transient over/undershoot, settling	No oscillation; bounded excursion
Start-up	V_{IN} ramp	V_{out} trajectory	Monotonic, no latch state
PSRR	AC on V_{IN}	V_{out}/V_{IN} (f)	Spec to be set from load needs

IX. LIMITATIONS AND FUTURE WORK

Observed limitations from the evidence: the 51.8° phase margin; the second-stage current and mirror-ratio deviation; the open-loop-only transient characterization; the unresolved power-measurement definition; and the standalone testbench operating point (ideal 1.8 V supply, output resting near 0.15 V). Anticipated — but not yet observed — challenges concern the system: start-up from a weak 0.3–1 V source, headroom for the amplifier and pass device when the stored rail sags, reference accuracy at low voltage, quiescent-current budget once all blocks are active, loop stability with the 10 µF/16 Ω output network, output ripple under duty-cycled loads, and the absence of corner, mismatch and post-layout verification. Table IX ranks these with remedies; the prioritized plan is: (1) confirm the power definition, $C_C/R_Z/C_L$ values and re-tune the compensation to ≥60°; (2) close and DC-verify the LDO; (3) loop-gain and transient verification with the output network; (4) implement the 0.5 V reference [6]; (5) design the conditioning/boost stage and storage [7]; (6) add the voltage monitor and load switch; (7) full-PMU simulation under varying input and load; (8) PVT, mismatch and post-layout checks as tools permit.

TABLE IX
LIMITATIONS AND FUTURE WORK

Issue	Technical impact	Proposed remedy	Priority
Phase margin 51.8° < 60° (observed)	Reduced damping; risk inside LDO loop	Increase C_C / retune R_Z; re-verify GBW	High
2nd-stage current −36 % vs design (observed)	Lower SR/A_{v2} than intended; power bookkeeping	Re-size M7 or accept and re-document bias	Medium
Power definition unconfirmed (observed)	4.73 vs 5.89 µW ambiguity	Re-measure with explicit $V_{DD}\cdot I$ expression	High
Open-loop transient only (observed)	No closed-loop settling evidence	Unity-gain and LDO load-step benches	High
Harvester headroom 0.3–1 V (anticipated)	Cannot feed 1.2 V LDO directly	Boost/SC stage + storage [7]	High
Low-voltage reference accuracy (anticipated)	Output set-point error via (7)	Dedicated reference design [6]	Medium
No PVT/mismatch/post-layout data (observed gap)	Unquantified spread	Corners, Monte Carlo, extraction	Medium

X. CONCLUSION

This work set out the architecture of a solar energy-harvesting PMU and completed, at schematic level in 0.18 µm CMOS, its central analog block: a two-stage, PMOS-input, Miller-compensated error amplifier intended for the PMU's LDO. Square-law hand sizing carried the design to a bias point that simulation confirms device by device, and the standalone characterization yields 70.5 dB gain, 1.629 MHz unity-gain bandwidth, 87.8 dB CMRR, ≈0.612 V/µs slew rate and microwatt-level power from a 1.8 V supply — figures compatible with its role as an LDO error amplifier. The characterization is equally clear about what is not yet achieved: the 51.8° phase margin must be raised before loop closure, the power-measurement definition needs confirmation, and the LDO, reference, conditioning stage, storage and monitor exist only as a specified plan with starting component values. The next milestone — closing the LDO around the unchanged amplifier and verifying regulation and loop stability against Table VIII — will convert the proposed PMU from an architecture into a demonstrated system.

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